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PROJECT TITLE	:**CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE**
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CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

SALESFORCE IMPLEMENTATION

User Story:

The Customer Support Ticket Priority Prediction and Automated Assignment System is a Salesforce solution that reads the description of a customer support ticket, decides whether its priority is High, Medium or Low, and starts the right backend automation. Support tickets are stored in the Support Ticket Intelligence custom object and are linked to Accounts and Contacts. An Auto-Launched Flow retrieves the latest ticket of an account, analyzes its description, sets the priority and, for High-priority tickets, creates an urgent handling task and assigns a senior support agent. An Agentforce subagent named Support Ticket Priority Analysis lets support users run this process through a simple conversation by giving an Account Name. Together, these features reduce manual triage, improve response time and give the support team clearer control over urgent issues.

1. INTRODUCTION

1.1 Project Overview

Customer support teams receive many tickets every day through email, web forms and chat. In many organizations, prioritizing and assigning these tickets is done by hand, which is slow, inconsistent and can hide urgent problems behind routine requests. This project provides a Salesforce-based solution that analyzes ticket information, classifies urgency, assigns the appropriate support level and starts automation for critical cases.

The solution models the business with one custom object (Support Ticket Intelligence) connected to the standard Account, Contact, User and Task objects. A record-triggered style logic is implemented in an Auto-Launched Flow, and Agentforce provides the conversational front end. The complete solution was configured and tested in a Salesforce Developer Edition org for academic purposes, following the practices of a real implementation.

1.2 Purpose

The system exists to make ticket handling faster, more consistent and easier to monitor. Specifically, it is designed to:

- Automatically classify support tickets as High, Medium or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce the manual effort of prioritizing and assigning tickets.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve ticket response and resolution efficiency.

1.3 Objectives and Business Value

The main objective is to give the support team a single, reliable and automated way to decide which tickets need attention first. The business value comes from:

- **Less manual work:** priority setting, task creation and assignment run automatically.
- **Faster response:** critical tickets are identified from their description without waiting for a person to read them.
- **Better assignment:** urgent tickets reach a senior support agent.
- **Visibility:** agents and managers can see ticket status, priority and SLA risk.
- **Easy access:** support users get a clear answer by chatting with Agentforce.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	prioritization can be manual	frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind lower-priority tickets	support teams handle large ticket volumes	concerned
PS-3	Support Manager	assign tickets to the right support level	assignment may need manual decisions	ticket urgency is not consistently analyzed	overwhelmed
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	descriptions contain different urgency signals	uncertain
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may go unnoticed	SLA risk needs additional checking	concerned

2.2 Empathy Map Canvas

User Persona: Support Agent

Primary responsibility: reviewing customer tickets, handling assigned cases and focusing on urgent issues.

Quadrant	Description
Think & Feel	Wants critical issues identified quickly so urgent customer problems receive attention.
Hear	Customer issue descriptions and support requests.
See	Ticket status, issue type, description, priority level, assignment and SLA risk.
Say & Do	Reviews tickets, handles assigned cases and focuses on urgent support issues.
Pain	Manual prioritization, missed urgency signals and delays in handling critical tickets.
Gain	Automatic priority classification, clearer assignment, urgent task creation and reduced manual effort.

2.3 Brainstorming and Idea Prioritization

Ideas Considered

Idea	Approach	Key Limitation
Manual ticket registers	Record ticket details on paper or in notebooks	Slow, error-prone and hard to track
Spreadsheet tracking	Track tickets in spreadsheets	Little automation and no AI support
Standalone support portal	Build a dedicated support portal	High development and maintenance effort
Salesforce CRM with Agentforce	Centralize tickets, automation and AI on Salesforce	Needs careful configuration

Final Selection: Salesforce CRM with Agentforce was selected for these reasons:

- **Automation:** Flow automates priority setting, task creation and assignment.
- **AI assistance:** Agentforce analyzes ticket details through conversation.
- **Scalability:** The platform grows with more tickets, users and requirements.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stage	Journey Stage	What Happens
1	Raise Ticket	The customer reports an issue through a support channel.
2	Ticket Received	The ticket is stored in the Support Ticket Intelligence object.
3	AI Analysis and Prioritization	Agentforce and the Flow read the description and classify priority.
4	Auto Assignment	The ticket is assigned to the right support level.
5	Urgent Task Creation	High-priority tickets get an Urgent Ticket Handling task.
6	Agent Works on Ticket	The agent investigates, updates the status and resolves the issue.
7	Customer Gets Updates	The customer is informed about progress and resolution.
8	Feedback and Improvement	Feedback and ticket data are used for continuous improvement.

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

Non-Functional Requirements

NFR	Description
Usability	Clear Agentforce interaction and understandable priority messages.
Reliability	Consistent priority classification based on configured conditions.
Performance	Quick retrieval and analysis of the latest support ticket.

NFR	Description
Security	Controlled Salesforce access to customer and ticket information.
Maintainability	Flow and Agentforce configuration can be updated easily.
Scalability	Supports growing ticket volumes through Salesforce automation.

3.3 Data Flow Diagram

Main Flow

Customer or Support User → Account Name → Salesforce Agentforce → Support Ticket Priority Analysis subagent → Auto-Launched Flow → Account → Latest Support Ticket → Ticket Description → Analyze Description → Priority Level → High / Medium / Low decision → Agent Assignment → High-Priority Task (if applicable) → Final Response.

Supporting Flows

- **Agentforce flow:** Support user gives the Account Name → Agentforce calls the Flow → ticket analysis → priority and assignment output → response to the user.
- **High-priority handling:** Ticket priority is High → Create Task (Urgent Ticket Handling) → assign Senior Support Agent → urgent handling message.
- **Medium and Low handling:** Ticket is Medium or Low → priority is set → message says it will be handled shortly or is queued.

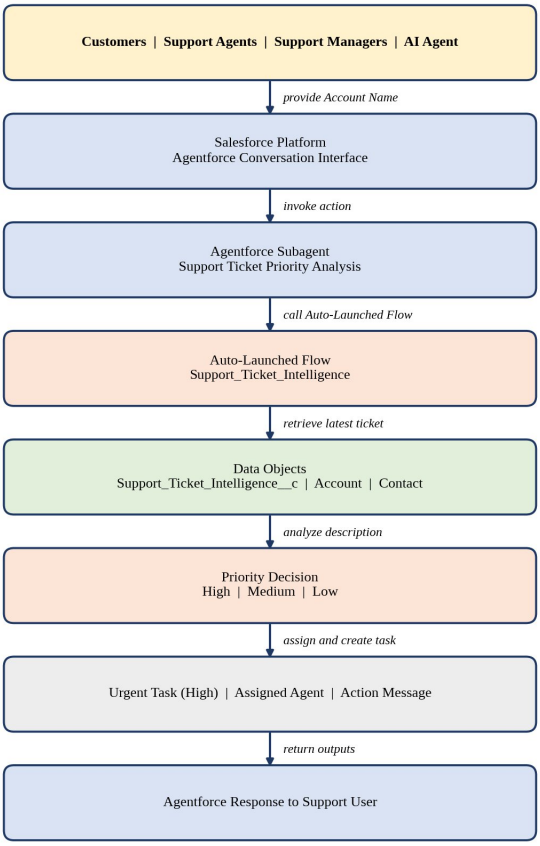


Figure 1: Data flow diagram of the Support Ticket Priority System

3.4 Technology Stack

Layer	Technology
Platform	Salesforce, Developer Edition
Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records
Security	Profiles, roles, permission sets, sharing settings, Einstein Trust Layer
Reporting	Reports and dashboards
Output	Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

Each problem identified in the ideation phase is matched with a Salesforce capability in the table below.

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes the description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	Keyword-based High-priority decision	Critical cases receive attention
Manual task creation	Flow creates a Task for High priority	Reduced manual effort
Manual support assignment	Flow sets the assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA risk decision	Better risk awareness
Need for conversational access	Agentforce subagent	AI-assisted interaction

4.2 Proposed Solution

Parameter	Description
Problem	Manual ticket prioritization and assignment delay the handling of critical customer issues.
Solution	A Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System.
Innovation	Agentforce AI with an Auto-Launched Flow that analyzes descriptions, classifies priority, assigns a support level and triggers urgent-task automation.
Impact	Faster identification of critical tickets, reduced manual effort, improved response time and better team productivity.
Business Model	Customer support operations involving customers, support agents, managers and AI-assisted ticket handling.
Scalability	Supports growing ticket volumes, with scope for more assignment rules, SLA monitoring and priority indicators.

4.3 Solution Architecture

The solution is made up of the following components:

- Custom object: Support Ticket Intelligence, with Account, Contact, User and Task records

- Auto-Launched Flow with decision, assignment and create-record elements
- Agentforce subagent and Flow-based agent action
- Profiles, roles, permission sets and sharing settings
- Lightning record page, reports and dashboard

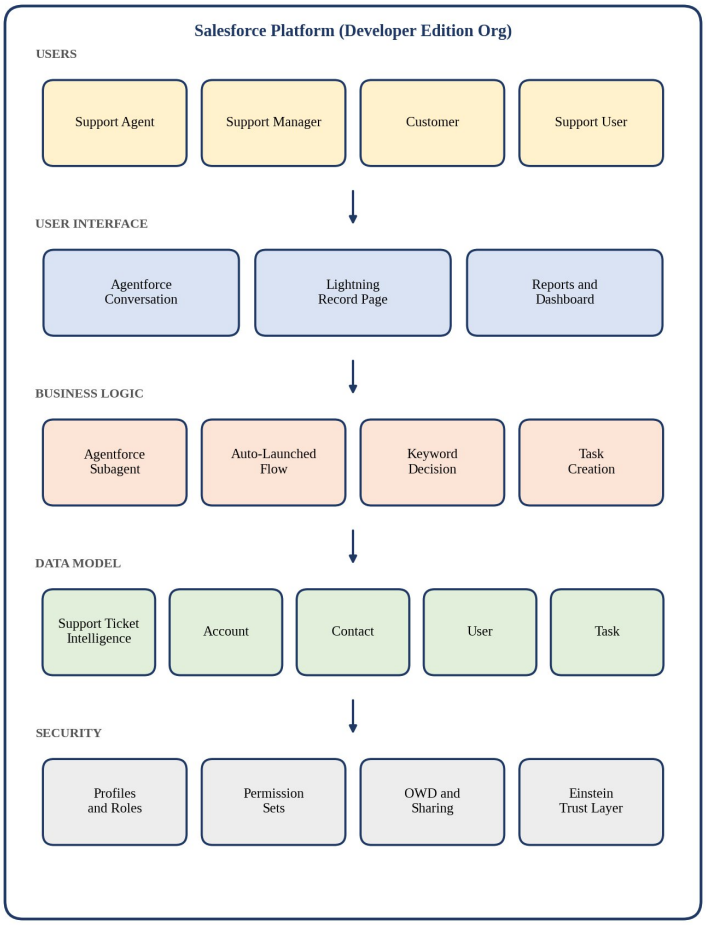


Figure 2: Solution architecture of the Support Ticket Priority System

5. PROJECT PLANNING AND SCHEDULING

5.1 Project Planning

The project followed an Agile approach: requirements were grouped into epics, split into user stories with story points, and delivered in six sprints. Velocity from each sprint was used to estimate the next.

Product Backlog and Sprint Schedule

Sprint	Epic	US No	User Story / Task	Points	Priority	Member
Sprint-1	Developer Setup	US-01	As a system administrator, I want to configure the Salesforce environment so that the solution can be developed.	3	High	Member 1
Sprint-2	Data Modeling	US-02	As an admin, I want to create the Support Ticket Intelligence object with fields and relationships so that tickets are stored systematically.	5	High	Member 2
Sprint-2	Data Modeling	US-03	As a support user, I want tickets linked to Accounts and Contacts so that the latest customer ticket can be identified.	3	High	Member 3
Sprint-3	Automation	US-04	As a support agent, I want a Flow to retrieve the latest ticket and analyze its description so that priority is determined automatically.	5	High	Member 4
Sprint-3	Automation	US-05	As a support agent, I want tickets classified as High, Medium or Low so that critical issues get attention first.	5	High	Member 4
Sprint-4	Agentforce and Automation	US-06	As an AI agent, I want to analyze ticket details and trigger the	8	High	Member 1

Sprint	Epic	US No	User Story / Task	Points	Priority	Member
			Flow so that manual prioritization is reduced.			
Sprint-4	Task and Assignment	US-07	As a support manager, I want High-priority tickets to create an urgent task and be assigned to the right level so that they are handled quickly.	5	High	Member 2
Sprint-5	SLA Management	US-08	As a support manager, I want SLA breach risk checked for older unresolved tickets so that issues are identified early.	5	Medium	Member 3
Sprint-6	Agentforce and Conversational Support	US-09	As a support user, I want to give an Account Name to Agentforce and receive the ticket ID, priority, assigned agent and action message.	5	Medium	Member 4

Project Tracker, Velocity and Burndown

Sprint	Total Points	Duration	Start Date (Planned)	End Date (Planned)	Points Completed	Release Date (Actual)
Sprint-1	3	6 days	DD Mon YYYY	DD Mon YYYY		
Sprint-2	8	6 days	DD Mon YYYY	DD Mon YYYY		
Sprint-3	10	6 days	DD Mon YYYY	DD Mon YYYY		
Sprint-4	13	6 days	DD Mon YYYY	DD Mon YYYY		
Sprint-5	5	6 days	DD Mon YYYY	DD Mon YYYY		
Sprint-6	5	6 days	DD Mon	DD Mon		

Sprint	Total Points	Duration	Start Date (Planned)	End Date (Planned)	Points Completed	Release Date (Actual)
			YYYY	YYYY		

Average velocity = total story points ÷ number of sprints = 44 ÷ 6 ≈ 7.3 points per sprint. Replace the date placeholders with your own sprint dates.

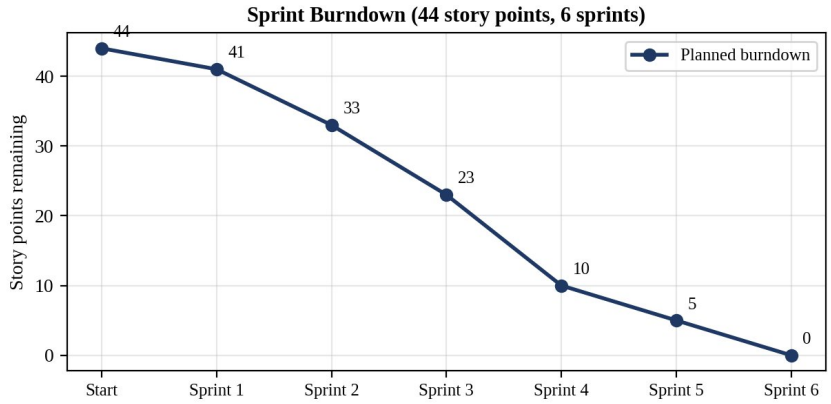


Figure 3: Planned sprint burndown chart

6. PROJECT DEVELOPMENT PHASE

Phase 1: Requirement Analysis and Planning

The goal of this phase was to define what the system must do and how it would be built. The system lets support users analyze the latest ticket of an account, classifies its priority from the description, creates tasks for urgent tickets, assigns a support level and returns a clear message through Agentforce.

Functional Scope

- Custom object Support Ticket Intelligence with fields for ticket details, priority, status, assigned agent and SLA risk.
- Relationships: lookups to Account, Contact and User, and Tasks related to the ticket.
- An Auto-Launched Flow for retrieval, keyword-based analysis, priority setting, task creation and assignment.
- An Agentforce subagent and a Flow-based action for conversational access.
- Profiles, roles and permission sets for user-level access.
- Reports and a dashboard for ticket priority and status insights.

Stakeholder Mapping

Stakeholder	Role in the System
Customers	Raise issues and expect quick resolution.
Support Agents	Handle tickets and need clear priority and assignment.
Support Managers	Monitor workload, ticket status and team performance.
AI Agent (Agentforce)	Automates analysis and triggers the required actions.

Execution Roadmap

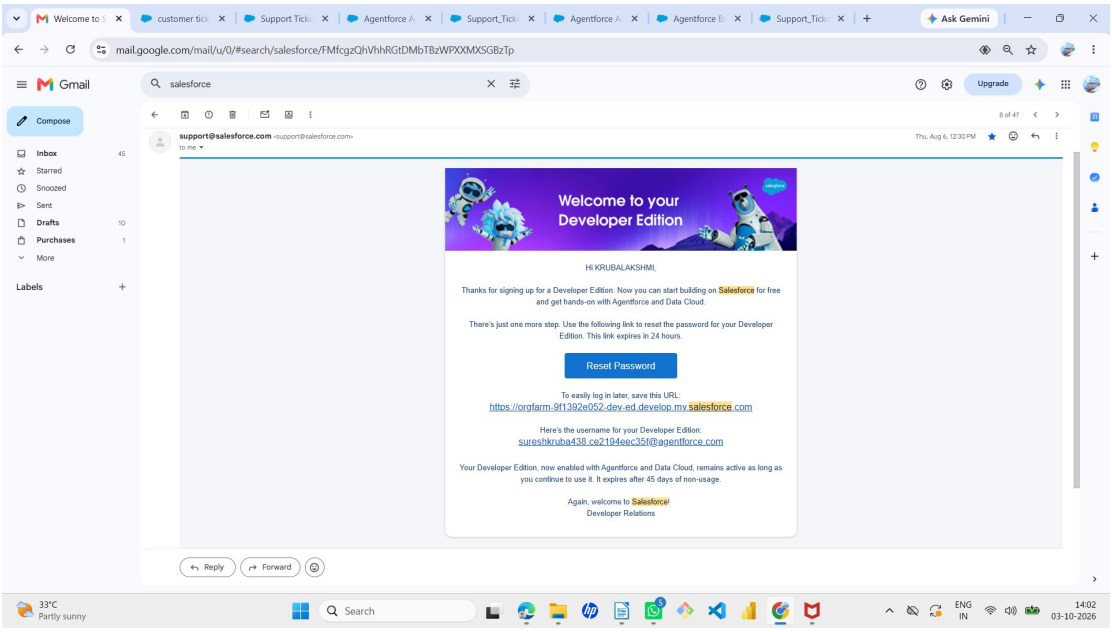
- Developer org setup and custom object creation.
- Fields, relationships and validation rules .
- Auto-Launched Flow for priority analysis and task creation.
- Agentforce subagent and Flow-based action.
- Lightning record page, reports and dashboard.
- Data migration, security setup, testing and documentation.

Phase 2: Backend Development and Configurations

This phase built the working core of the system. Declarative tools (custom object, fields, validation rule and Flow) hold and process the ticket data, while Agentforce adds the conversational layer. Configuration and logic were created and tested step by step in the Developer org.

Milestone 1: Salesforce Developer Account

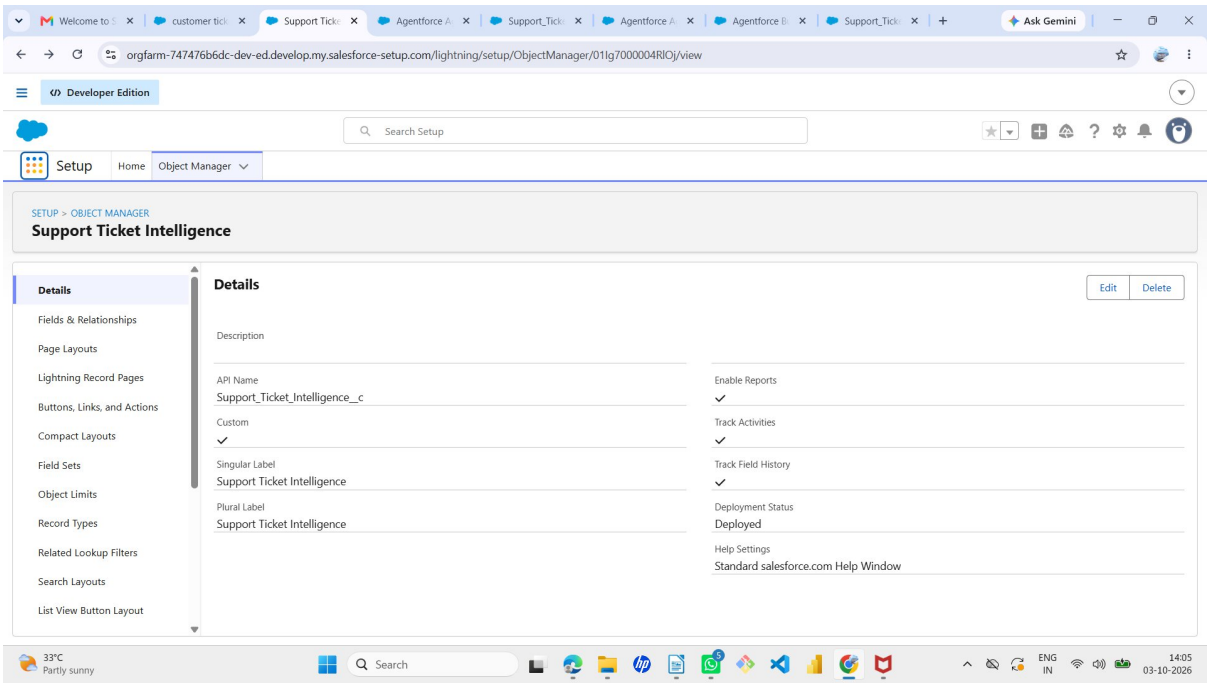
- Open the Salesforce Developer Edition sign-up page and enter your name, email, role (Developer), company (college name), country, postal code and a unique username in the form name@organization.com.
- Open the verification email, click Verify Account, set a password and answer the security question to reach the Setup home page.



Milestone 2: Custom Object

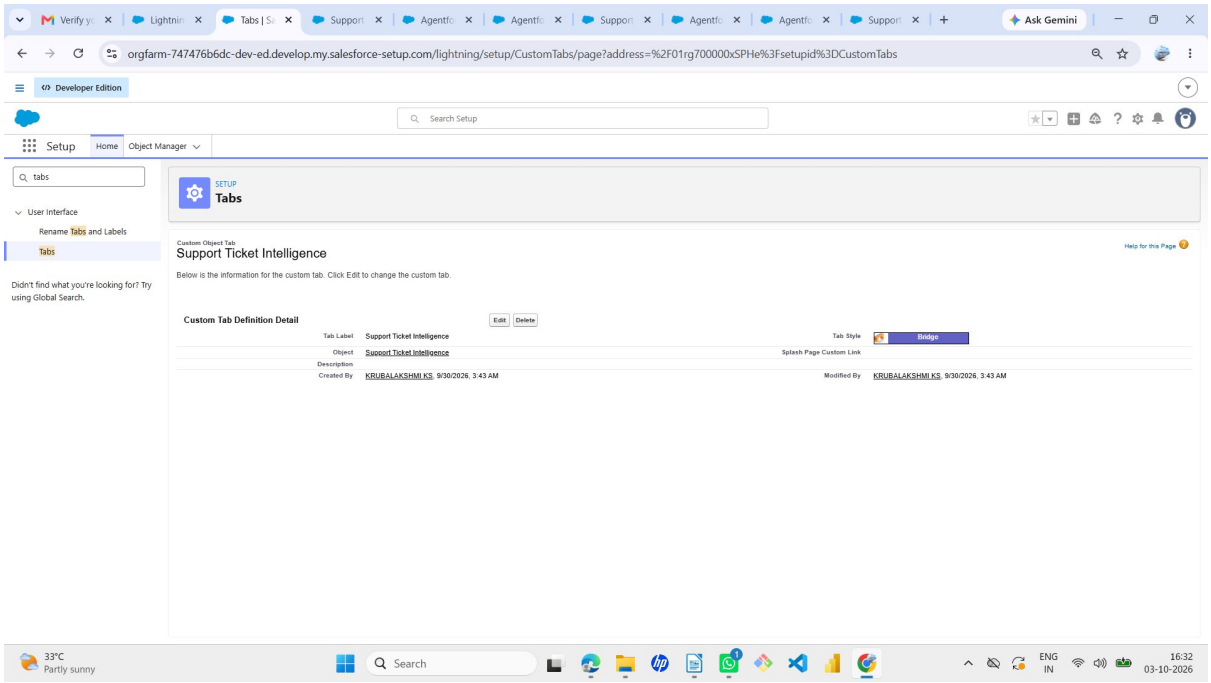
An object is a database table that stores organization-specific data. The custom object was created from Setup → Object Manager → Create → Custom Object.

Object	API Name	Purpose
Support Ticket Intelligence	Support_Ticket_Intelligence__c	Stores each support ticket with its priority, status and assignment



Milestone 3: Tab

A custom object tab was created for Support Ticket Intelligence (Setup → Tabs → New under Custom Object Tabs), with a tab style selected and the other options left at their defaults.



Milestone 4: Fields and Relationships

Standard fields exist on every object; custom fields were added to meet the business requirements.

Field Label	API Name	Data Type	Description / Values
Ticket Number	Ticket_Number__c	Auto Number	Format TKT-{0000}
Customer	Customer__c	Lookup (Account)	Related account
Contact	Contact__c	Lookup (Contact)	Customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken

Relationships

Relationship	Type
Support Ticket Intelligence → Account (Customer)	Lookup
Support Ticket Intelligence → Contact	Lookup
Support Ticket Intelligence → User (Assigned To)	Lookup
Task → Support Ticket Intelligence (WhatId)	Related to (activity)

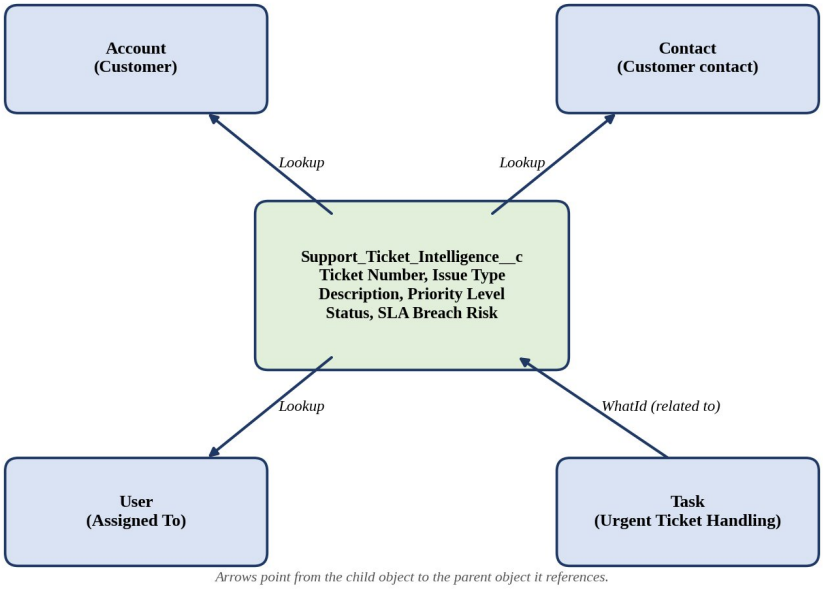
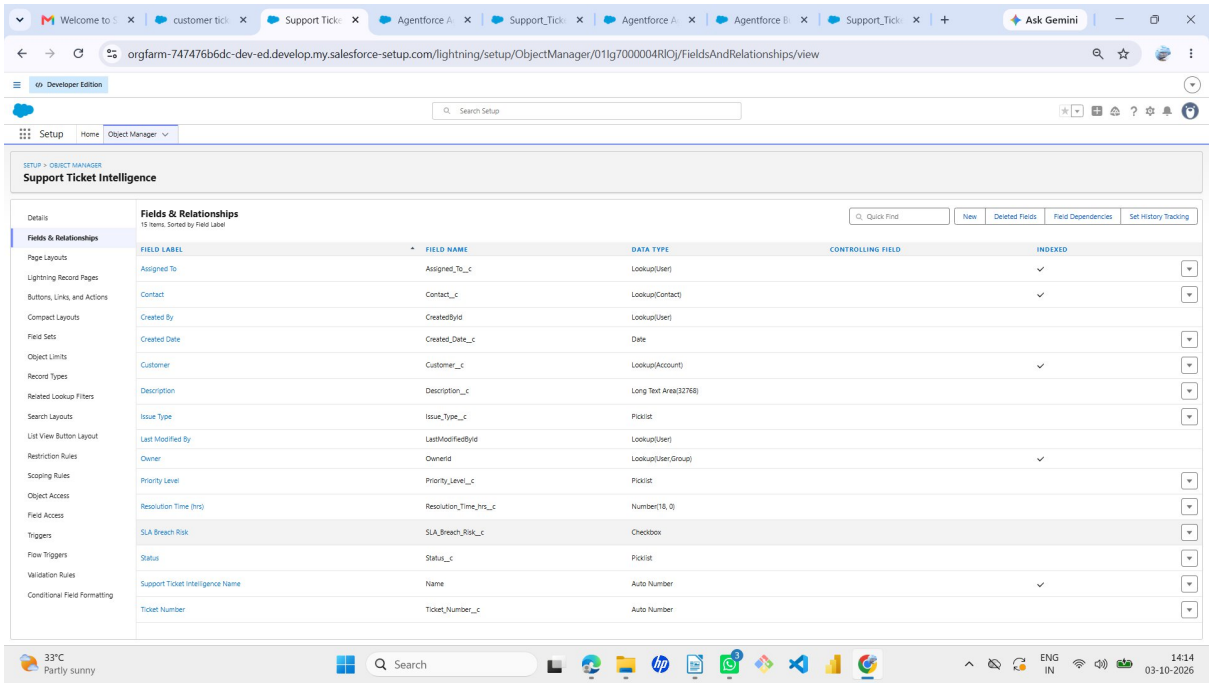


Figure 4: Data model: objects and relationships



Milestone 5: Validation Rule

A validation rule checks data when a record is saved and stops the save if the condition is not met. The rule Description_Required was created on Support_Ticket_Intelligence__c so that every ticket carries a description for the priority analysis.

```
Rule Name:   Description_Required
Active:      Yes
Formula:     ISBLANK(Description__c)
Message:     Please enter the ticket description.
```
Location: Field → Description

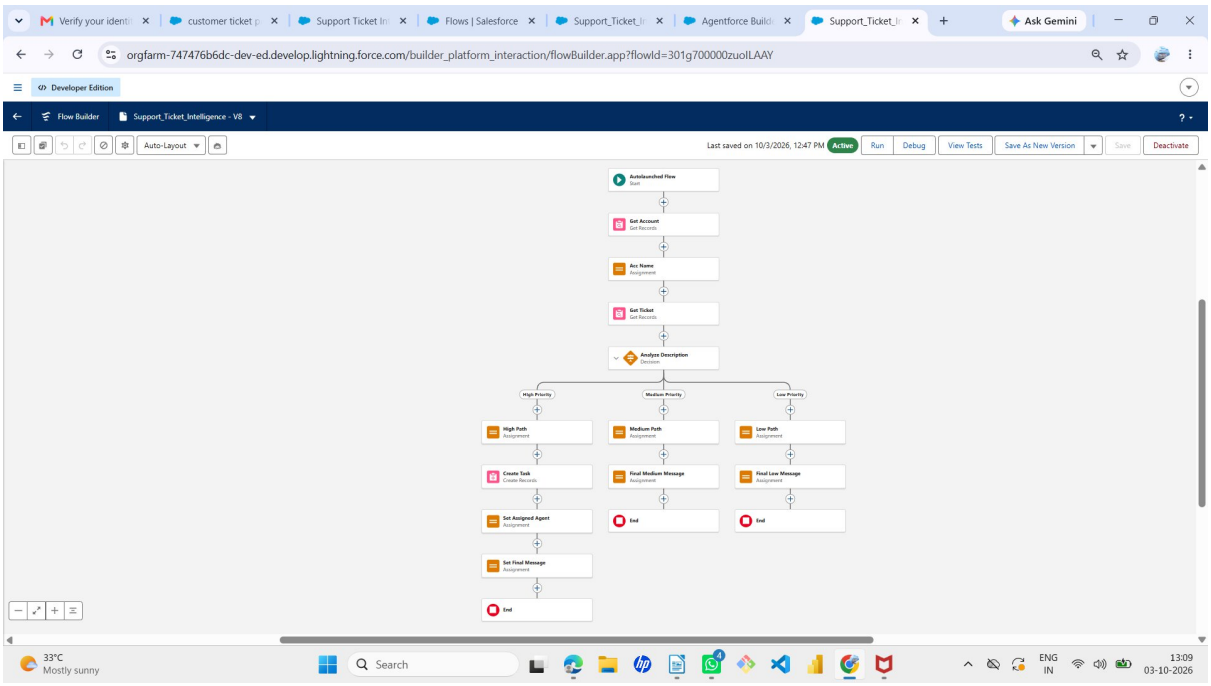
Milestone 6: Auto-Launched Flow

Scenario: when the Flow receives an Account Name, it finds the latest ticket of that account, analyzes the description and sets the priority. For High priority it creates an urgent task and assigns a senior support agent. The Flow is called by the Agentforce action and has no trigger of its own.

Variable Label	API Name	Type	Direction
Account Name	varAccountName	Text	Input
Account Id	varAccountId	Text	Output
Ticket Id	varTicketId	Text	Output
Priority Level	varPriorityLevel	Text	Output
Assigned Agent	varAssignedTo	Text	Output
Action Message	varActionMessage	Text	Output

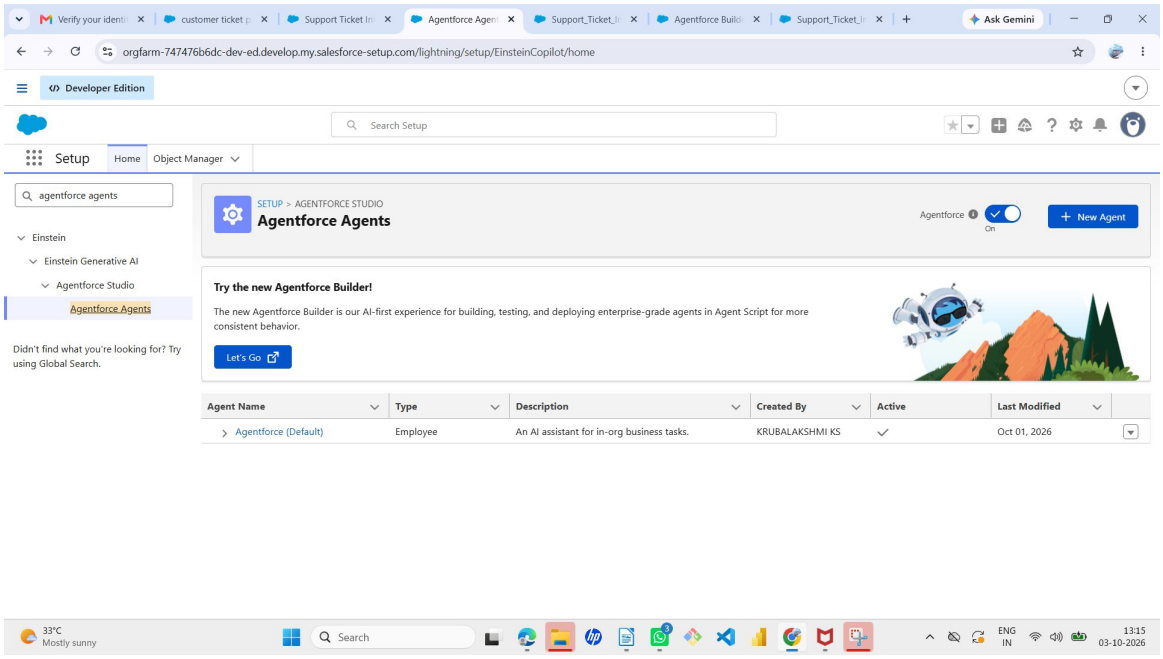
Step	Element	Label	Configuration
1	Get Records	Get Account	Account where Name equals varAccountName; sort by CreatedDate descending; first record only.
2	Assignment	Store Account Id	varAccountId = Get_Account.Id
3	Get Records	Get Ticket	Support_Ticket_Intelligence__c where Customer__c equals varAccountId; latest first; first record only.
4	Assignment	Store Ticket Id	varTicketId = Get_Ticket.Id
5	Decision	Analyze Description	High: description contains urgent, not working or failure. Medium: contains issue, slow or delay. Default outcome: Low.
6	Assignment	Set Priority Level	varPriorityLevel = High, Medium or Low on the matching path.
7	Create Records	Create Task (High only)	Subject Urgent Ticket Handling; WhatId = varTicketId; Priority High; Status Not Started.
8	Assignment	Set Assigned Agent	varAssignedTo = Senior Support Agent (High path).
9	Decision	SLA Risk (optional)	Checks whether the ticket is older than two days.
10	Assignment	Set Final Message	High: urgent ticket assigned to

Step	Element	Label	Configuration
			senior agent. Medium: will be handled shortly. Low: queued for processing.
11	Save and Activate	Support_Ticket_Intelligence	Save the Flow and activate it.



Milestone 7: Enabling Agentforce

1. Click the gear icon and open Salesforce Go.
2. Search for Agentforce (Default), select it and choose Get Started, then Turn On and Confirm.
3. Search Agents in the Setup Quick Find box, make sure Agentforce is enabled, and open Agentforce (Default) in Builder.



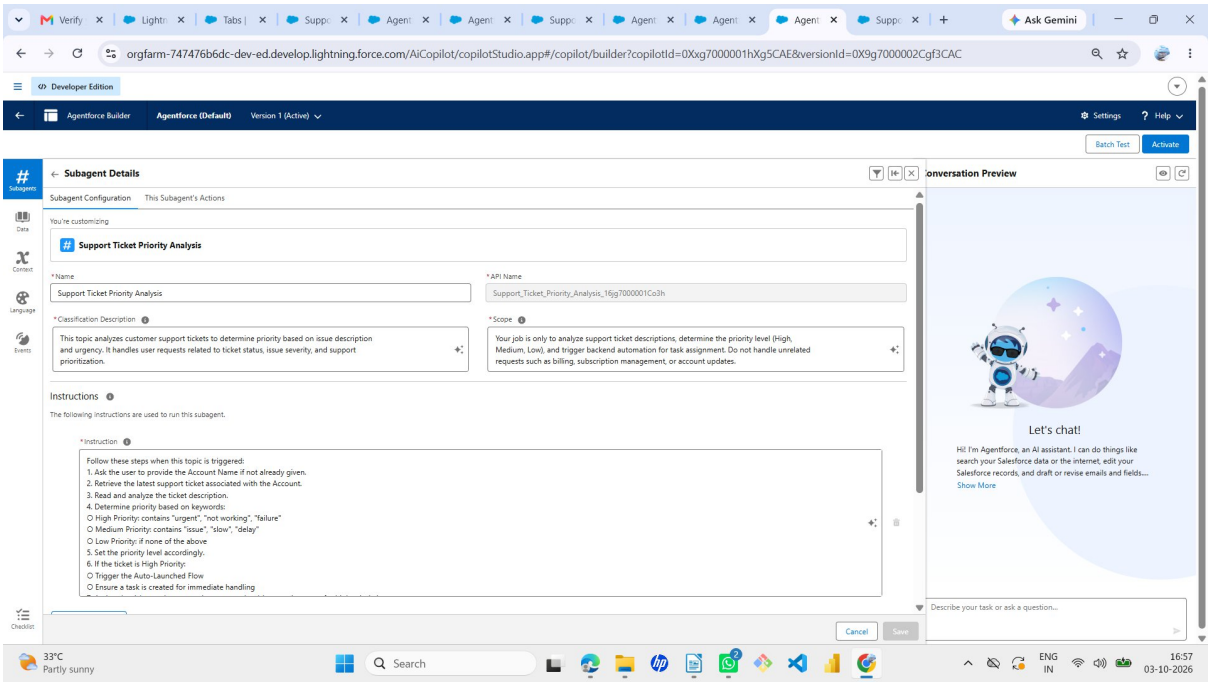
Milestone 8: Subagent Configuration

In Agentforce Builder, under Subagents, choose New → New Subagent and enter the details below.

Setting	Value
Name	Support Ticket Priority Analysis
API Name	Support_Ticket_Priority_Analysis
Classification Description	Analyzes customer support tickets to determine priority based on issue description and urgency; handles requests about ticket status, issue severity and support prioritization.
Scope	Only analyze support ticket descriptions, determine the priority level (High, Medium, Low) and trigger backend automation for task assignment. Do not handle billing, subscription management or account updates.

Instructions given to the subagent:

1. Ask the user for the Account Name if it is not already given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority: High if it contains urgent, not working or failure; Medium if it contains issue, slow or delay; otherwise Low.
5. Set the priority level accordingly.
6. For High priority, trigger the Auto-Launched Flow and ensure an urgent task is created.
7. Assign the ticket to the appropriate support level, such as a senior agent for High priority.
8. Return a clear message: High means urgent handling is initiated, Medium means it will be handled shortly, Low means it is queued.
9. Do not update records directly or send emails; use the Flow for all actions.



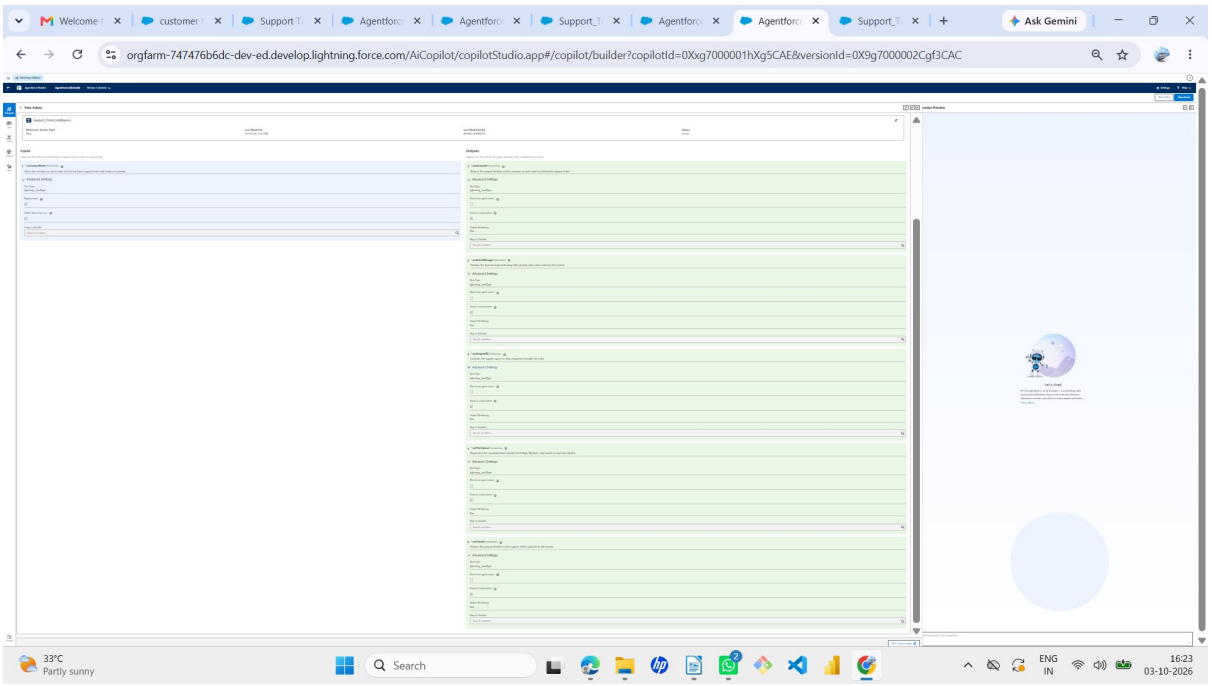
Milestone 9: Agent Action

Under This Topic's Actions, choose New → Create New Action, set the Reference Action Type to Flow and select the Support Ticket Intelligence Flow. Then configure the action as follows.

Action Field	Name	Description
Action Description	-	Analyzes support tickets for the provided

Action Field	Name	Description
		account name, determines priority from the issue description and triggers automation for High-priority cases.
Loading Text	-	Analyzing ticket details and determining priority...
Input	varAccountName	Account name used to fetch the latest ticket (required; collected from the user).
Output	varAccountId	Unique ID of the account used for retrieval.
Output	varTicketId	Unique ID of the analyzed ticket.
Output	varPriorityLevel	Calculated priority: High, Medium or Low.
Output	varAssignedTo	Support agent or team assigned.
Output	varActionMessage	Final message describing priority and actions taken.

All outputs are set to Show in Conversation with Text rendering.

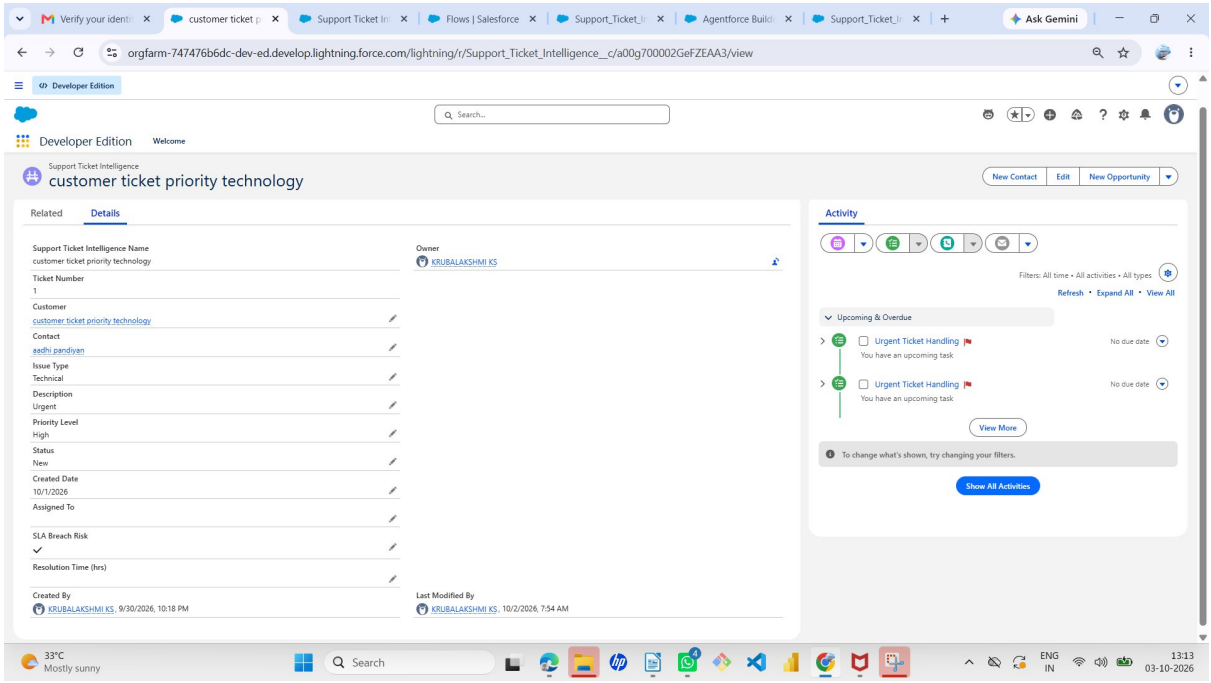


Phase 3: UI/UX Development and Customization

This phase gave users a simple workspace. Support users work with ticket records on a Lightning record page, interact with the Agentforce conversation panel, and view reports and a dashboard that turn ticket data into visual summaries.

Milestone 10: Lightning Record Page

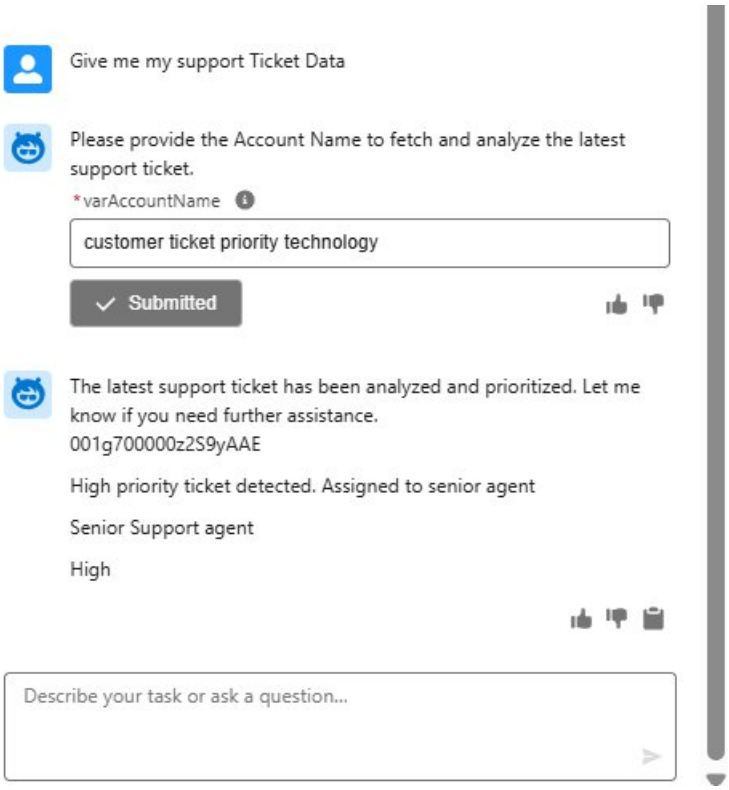
The record page of Support Ticket Intelligence shows the customer, contact, issue type, description, priority level and status. The Activity timeline lists the Urgent Ticket Handling tasks created automatically by the Flow for High-priority tickets.



Milestone 11: Agentforce Conversation

Support users test and use the solution in the Conversation Preview. The agent asks for the Account Name, runs the action, and returns the ticket ID, priority, assigned agent and action message.

8.



ADVANTAGES AND DISADVANTAGES

Advantages

- **Automatic prioritization:** tickets are classified as High, Medium or Low from their description.
- **Faster handling of critical issues:** agents focus on urgent tickets first.
- **Reduced manual effort:** analysis, priority setting, task creation and assignment are automated.
- **Urgent task creation:** High-priority tickets automatically get an Urgent Ticket Handling task.
- **Better assignment:** urgent tickets reach a senior support agent.
- **Conversational support:** users get results by chatting with Agentforce.
- **SLA awareness:** older unresolved tickets can be checked for breach risk.
- **Scalability:** Salesforce automation supports growing ticket volumes.
- **Higher productivity:** teams spend more time resolving and less time sorting.

Disadvantages

- **Keyword-based logic:** priority depends on configured keywords such as urgent, slow and delay.
- **Limited context understanding:** some urgency signals may be missed because the logic uses fixed conditions.
- **Dependence on data quality:** correct Account, Contact and ticket data is needed.

- **Configuration dependency:** the Flow and the subagent instructions must be maintained properly.
- **Limited scope:** the subagent handles only ticket analysis, not billing or subscriptions.
- **Basic SLA monitoring:** the SLA check is optional and not a full escalation solution.
- **Developer Edition limits:** the project was built for academic purposes and is not deployed to production.
- **Limited analytics:** advanced trend analysis is part of future scope.

9. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System shows how Salesforce Flow and Agentforce can work together to reduce manual ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, assigns a support level and returns a clear conversational response.

Building the project covered the full implementation cycle: designing a data model, writing a validation rule and a Flow, configuring an Agentforce subagent and action, securing data with profiles, roles, permission sets and sharing settings, migrating data, testing and documenting the result. Although the solution was built and tested in a Developer Edition org without a production deployment, it follows real-world Salesforce practice and provides a base for future growth.

10. FUTURE SCOPE

- Add more priority indicators and business rules, including sentiment analysis.
- Extend Agentforce outputs with additional ticket information.
- Add advanced SLA monitoring with escalation and notifications.
- Support more teams, skills and workload-based assignment through Omni-Channel.
- Introduce analytics for priority trends and resolution performance.
- Suggest automatic replies to customers.
- Production deployment using sandboxes, change sets or Salesforce DX.

11. APPENDIX

- **A. Custom Object:** Support Ticket Intelligence (Support_Ticket_Intelligence__c).
- **B. Key Salesforce Records:** Account, Contact, User and Task.
- **C. Automation:** Support_Ticket_Intelligence Auto-Launched Flow and the Description_Required validation rule.
- **D. Agentforce:** Support Ticket Priority Analysis subagent and the Flow-based Support Ticket Intelligence action.
- **E. Security:** Profiles, role hierarchy, Ticket Automation Access permission set and sharing settings.
- **F. Analytics:** Tickets by Priority report and Support Operations Dashboard.